

Integrated Action, Kinetic Bound, and the Weighted Metric Trace Lemma

(Route δ , Plan 2, Building Block **WEIGHTEDMETRICTRACE**)

Abstract

This building block chains conditional, explicitly named ingredients into a pointwise metric distance bound at the deficit endpoint $\hat{\rho} := \rho_\delta = 1 - \kappa\sqrt{\delta}$, where throughout this note we write $\delta := \delta_T(\Omega)$. The repaired version does not use the unresolved centroid space-time Π estimate. Instead, on the levels where the isoperimetric defect is small it assumes a same-centre Fusco–Julin trace package: one bounded Borel centre controls both the Fraenkel distance and the homothetic trace. On the same good levels, the low-gradient part of $\partial^* E_\rho$ is self-truncated and charged by the Cauchy–Schwarz defect D_H via [5], replacing the former pointwise lower-gradient hypothesis. The metric first-variation and all bad-level kinetic estimates are kept as conditional inputs, because the current upstream metric-framework finite-perimeter closure and the old uniform Lipschitz bound (A0) are not available as proved results. Under these inputs this yields the integrated action bound $\int (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu \leq C \delta_T$, which encodes the integrated isoperimetric defect D_I (via Fusco–Julin) in its distance term and the integrated Cauchy–Schwarz defect D_H (via the conditional metric first variation of [2]) in its kinetic term. A Talenti bad-set decomposition then gives the kinetic bound (K) in Lebesgue form. The abstract weighted metric trace lemma combines these two inputs — Markov on the distance term (encoding D_I) and Cauchy–Schwarz on the kinetic term (encoding D_H) — on the *order-one* window $J^* := [\rho_\delta - L^*, \rho_\delta]$ with $L^* := (\rho_\delta - \rho_*)/4$, to deliver the *uniform* pointwise bound $\sup_{\rho \in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C_* \delta_T$. In particular, setting $\hat{\rho} := \rho_\delta$, $d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C_* \delta_T$ at the upper endpoint of the deficit window. Either input alone would force a $\sqrt{\delta}$ exponent loss; together they yield the sharp bound at ρ_δ . The chain uses no pointwise (R)/(1.2) and no centroid Π -closure input.

1 Statements

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, $\kappa > 0$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $u = u_\Omega$ the variational torsion function, $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*, \kappa)$. We always take

$$\delta_0 \leq \left(\frac{1 - \rho_*}{2\kappa} \right)^2, \quad (1)$$

so that

$$\rho_\delta := 1 - \kappa\sqrt{\delta} \geq \frac{1 + \rho_*}{2}, \quad L^* := \frac{\rho_\delta - \rho_*}{4} \geq \frac{1 - \rho_*}{8}.$$

Thus the trace window used below is non-empty and has order-one length. Set $E_\rho := \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $w(\rho) := -t'(\rho) \geq 0$, and $\mu(d\rho) := w(\rho)d\rho$. Define $F_\rho := \rho^{-1}E_\rho \subset B_{R/\rho_*}$, $\gamma(\rho) := [F_\rho] \in \mathcal{X}$. For a.e. ρ , E_ρ has finite perimeter: this follows from the coarea formula applied to $u \in W_0^{1,2}(\Omega)$.

Theorem 1.1 (Conditional integrated action bound).

$$\int_{\rho_*}^{\rho_\delta} (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu(\rho) \leq C(n, R, \rho_*, \kappa) \delta_T(\Omega). \quad (2)$$

Theorem 1.2 (Conditional kinetic bound (K), Lebesgue form).

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C'(n, R, \rho_*, \kappa) \delta_T(\Omega). \quad (3)$$

Theorem 1.3 (Weighted metric trace). *Set $L^* := (\rho_\delta - \rho_*)/4$ and $J^* := [\rho_\delta - L^*, \rho_\delta]$. There exists $C_* = C_*(n, R, \rho_*, \kappa)$ such that*

$$\sup_{\rho \in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C_* \delta_T(\Omega). \quad (4)$$

In particular, setting $\hat{\rho} := \rho_\delta$,

$$d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C_* \delta_T(\Omega), \quad (5)$$

i.e. the metric distance to the ball is sharply controlled at the upper endpoint of the deficit window.

1.1 Self-truncated velocity input

The metric-derivative bound used below does *not* require a pointwise lower bound on $|\nabla u|$. Instead we use the following levelwise consequence of [5]. On every Fusco–Julin good level, i.e. on levels where $D_I(t(\rho)) \leq \eta_I$, the perimeter is uniformly bounded. Writing $f = |\nabla u|$ on $\partial^* E_\rho$ and $\bar{f}_\rho = |E_\rho|/P(E_\rho)$, split

$$\partial^* E_\rho = \{f \geq \bar{f}_\rho/2\} \cup \{f < \bar{f}_\rho/2\}.$$

The low-gradient part satisfies

$$\int_{\{f < \bar{f}_\rho/2\}} \frac{1}{f} d\mathcal{H}^{n-1} \leq C(n, \rho_*) D_H(t(\rho)),$$

by the weighted variance identity for D_H . On the complementary part, $f \geq \bar{f}_\rho/2$ gives the reciprocal-variance comparison needed in the old Hyp-G proof. Combining the two pieces yields

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C_{\text{tr}}(n, \rho_*) D_H(t(\rho)) \quad (\rho \in G_I). \quad (6)$$

Here

$$\bar{V}_\rho := \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

Thus tiny components or other anomalous low-gradient pieces are not controlled by a uniform pointwise constant; they are discarded at the level surface and paid for by D_H itself.

2 Levelwise quantitative input

For a.e. ρ write

$$\mathcal{I}(\rho) := D_I(t(\rho)), \quad \mathcal{H}(\rho) := D_H(t(\rho)), \quad \Delta P_\rho := P(E_\rho) - P(B_\rho).$$

Since $P(B_\rho) \geq n\omega_n \rho_*^{n-1}$ and $D_I(t(\rho)) = (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho))$,

$$0 \leq \Delta P_\rho \leq C_{n, \rho_*} \mathcal{I}(\rho). \quad (7)$$

The exact level-set deficit identity of [1, Eq. (eq:main), Thm. 5.1], together with the normalisation $|\Omega| = \omega_n$, $\delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0$, reads

$$\frac{2\delta_T(\Omega)}{\omega_n} = \frac{1}{\omega_n} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt.$$

On the deficit window $[\rho_*, \rho_\delta] \subset [0, 1]$, the weight $1/(n^2 \omega_n^{2/n} m(t)^{1-2/n})$ is bounded below by $c_{n, \rho_*} > 0$ since $m(t) \leq \omega_n$. Therefore

$$\int_{\rho_*}^{\rho_\delta} (\mathcal{H}(\rho) + \mathcal{I}(\rho)) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega), \quad (8)$$

where $C(n, \rho_*) := 2/c_{n, \rho_*} = 2n^2 \omega_n^{2/n+1} \rho_*^{n-2}$ for $n \geq 2$.

Remark 2.1 (Sign in the ρ change of variables). Because the volume radius increases as the threshold decreases, $t'(\rho) \leq 0$ and

$$dt = t'(\rho) d\rho, \quad d\mu(\rho) = (-t'(\rho)) d\rho = -dt.$$

Consequently, for a non-negative level integrand q ,

$$\int_{\rho_*}^{\rho_\delta} q(t(\rho)) d\mu(\rho) = \int_{t(\rho_\delta)}^{t(\rho_*)} q(t) dt.$$

This is the orientation used in (7); the older shorthand $dt = d\mu$ had the wrong sign.

Hypothesis 2.2 (Same-centre FJ trace package). There are constants $\eta_I > 0$, $C = C(n, R, \rho_*)$ and a bounded Borel centre field $\rho \mapsto z_\rho$ such that, on the good set

$$G_I := \{\rho \in [\rho_*, \rho_\delta] : \mathcal{I}(\rho) \leq \eta_I\},$$

the same centre z_ρ satisfies

$$|E_\rho \Delta B_\rho(z_\rho)|^2 \leq C \mathcal{I}(\rho), \quad (9)$$

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C \mathcal{I}(\rho), \quad (10)$$

where

$$\bar{V}_\rho := \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

This is not imported here from the present FJ bridge: it is the exact same-centre package still needed upstream. In particular, a centre that only minimizes Fraenkel asymmetry, or only minimizes normal oscillation, is not sufficient for this block.

Hypothesis 2.3 (Good-level metric upper gradient). For the centres in Hypothesis ??, the rescaled curve is absolutely continuous on $[\rho_*, \rho_\delta]$ and, for a.e. $\rho \in G_I$,

$$|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n, \rho_*} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

This is the finite-perimeter upper-gradient input requested from [2]; the unresolved lower-semicontinuity issue is not hidden in the proof below.

Hypothesis 2.4 (Bad-level kinetic inputs). With $B_I := [\rho_*, \rho_\delta] \setminus G_I$ and

$$B_{\tau_0} := \{\rho \in [\rho_*, \rho_\delta] : w(\rho) < \tau_0\}, \quad \tau_0 := \frac{\rho_*}{2n},$$

assume the two bad-level action bounds

$$\int_{B_I} |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu(\rho) \leq C(n, R, \rho_*) \delta, \quad (11)$$

$$\int_{B_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C(n, R, \rho_*) \delta. \quad (12)$$

These are the only bad-level kinetic estimates used below. They replace the old unsupported uniform Lipschitz assertion (A0).

Hypothesis 2.5 (Profile-gap and Talenti density inputs). The density $w = -t'$ satisfies the Talenti upper bound

$$0 \leq w(\rho) \leq \frac{1}{n} \quad \text{for a.e. } \rho \in [\rho_*, \rho_\delta],$$

and the profile-gap bad-set estimate

$$\mathcal{L}^1(B_{\tau_0}) \leq C(n, \rho_*) \delta. \quad (13)$$

The estimate (??) is the missing integrated Talenti/profile-gap input from [1].

3 The integrated action bound (5.2-int)

Proof of Theorem 1.1. Let $B_I := [\rho_*, \rho_\delta] \setminus G_I$. By (7),

$$\mu(B_I) \leq \eta_I^{-1} \int_{\rho_*}^{\rho_\delta} \mathcal{I}(\rho) d\mu(\rho) \leq C \delta_T(\Omega). \quad (14)$$

Distance term. On G_I , the quotient metric and the rescaling give

$$d_{\mathcal{X}}(F_\rho, B_1) \leq \rho^{-n} |E_\rho \Delta B_\rho(z_\rho)|, \quad (15)$$

hence (8) implies $d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C \mathcal{I}(\rho)$. On B_I we use only the trivial bound $d_{\mathcal{X}}(F_\rho, B_1) \leq 2\omega_n$, because both sets have volume ω_n . Therefore, by (7) and (12),

$$\int_{\rho_*}^{\rho_\delta} d_{\mathcal{X}}(F_\rho, B_1)^2 d\mu(\rho) \leq C \delta_T(\Omega). \quad (16)$$

Metric-derivative term. On G_I , Hypothesis ?? gives the per- ρ velocity bound

$$|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n, \rho_*} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

We specialise to $a = 0$ (a legitimate upper bound), then apply the triangle inequality $|V_\rho - H_{z_\rho, \rho}| \leq |V_\rho - \bar{V}_\rho| + |H_{z_\rho, \rho} - \bar{V}_\rho|$ under the integral, followed by $(a + b)^2 \leq 2a^2 + 2b^2$, to obtain

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq 2C_{n, \rho_*}^2 \left[\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| \right)^2 + \left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| \right)^2 \right]. \quad (17)$$

The first square is controlled by the self-truncated velocity estimate (5) from [5]:

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C_{\text{tr}}(n, \rho_*) \mathcal{H}(\rho), \quad (18)$$

and the second square is (11). Thus

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(n, R, \rho_*) (\mathcal{H}(\rho) + \mathcal{I}(\rho)) \quad \text{for a.e. } \rho \in G_I,$$

where $C(n, R, \rho_*)$ is polynomial in the constants of the self-truncated velocity estimate and of the Fusco–Julin package. On B_I we use only the conditional bad-isoperimetric-level action input (??). Combining this with (7) gives

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu(\rho) \leq C \delta_T(\Omega). \quad (19)$$

Equations (14) and (17) prove (1). $\square$

Remark 3.1. The good-level estimate is genuinely quantitative: the only square-root loss is the Fraenkel term from the same-centre FJ input, and it is squared before integration. The large-isoperimetric-defect levels are not estimated pointwise in this file; their distance contribution is paid for by $\mu(B_I) \leq C\delta$, while their kinetic contribution is exactly the conditional input (??).

4 The kinetic bound (K)

Remark 4.1 (No use of the old A0). The previous draft asserted a perimeter-free uniform metric Lipschitz bound (A0) from nestedness and bounded support. The audit found this unsupported. The Lebesgue kinetic estimate below therefore uses only the weighted action already proved and the explicit bad-density kinetic input (??).

Proof of Theorem 1.2. Split $[\rho_*, \rho_\delta] = G_{\tau_0} \sqcup B_{\tau_0}$ with $G_{\tau_0} := \{\rho : w(\rho) \geq \tau_0\}$ and $\tau_0 = \rho_*/(2n)$. On G_{τ_0} ,

$$\int_{G_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq \tau_0^{-1} \int_{G_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu(\rho) \leq C(n, \rho_*)\delta$$

by Theorem 1.1. On B_{τ_0} use the conditional bad-density action input (??). Adding gives (2). $\square$

5 The weighted metric trace lemma

Theorem 5.1 (Abstract weighted metric trace). *Let $\mu = w(\rho)d\rho$ on $[\rho_*, \rho_\delta]$. Assume:*

- (H1) $\int_{\rho_*}^{\rho_\delta} (d_{\mathcal{X}}(\gamma(\rho), B_1)^2 + |\dot{\gamma}|^2) d\mu \leq C\delta$,
- (H2) $\mu([a, b]) \geq c(b - a) - C_{\text{low}}\delta$ for every $[a, b] \subset [\rho_*, \rho_\delta]$,
- (H3) $0 \leq w(\rho) \leq M$ for a.e. ρ ,
- (H4) $\int_{\rho_*}^{\rho_\delta} |\dot{\gamma}|^2 d\rho \leq C'\delta$.

Set $L^* := (\rho_\delta - \rho_*)/4$ and $J^* := [\rho_\delta - L^*, \rho_\delta]$. Then

$$\sup_{\rho \in J^*} d_{\mathcal{X}}(\gamma(\rho), B_1)^2 \leq C_* \delta,$$

provided $\delta \leq cL^*/(2C_{\text{low}})$. Here $C_* := 2K_1 + 2L^*C'$ with $K_1 := 4C/(cL^*)$. In particular at the endpoint, $d_{\mathcal{X}}(\gamma(\rho_\delta), B_1)^2 \leq C_*\delta$.

Proof. Step 1: macroscopic Markov on the order-one window $J^ = [\rho_\delta - L^*, \rho_\delta]$.* By (H2), $\mu(J^*) \geq cL^* - C_{\text{low}}\delta \geq cL^*/2$. By (H1) and Chebyshev (Markov on the square),

$$\mu(\{\rho \in J^* : d_{\mathcal{X}}(\gamma(\rho), B_1)^2 > K_1\delta\}) \leq \frac{C\delta}{K_1\delta} < \mu(J^*)$$

for $K_1 := 4C/(cL^*)$ (any choice with $K_1 > 2C/\mu(J^*)$ works). Hence there exists $\rho_0 \in J^*$ with $d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 \leq K_1\delta$.

Step 2: kinetic transport across J^ .* For any $\rho \in J^*$, $|\rho - \rho_0| \leq L^*$ (the diameter of J^*). AGS Thm 1.1.2 [6, Thm. 1.1.2] gives $d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\rho)) \leq \int_{\rho_0}^{\rho} |\dot{\gamma}(s)| ds$; Cauchy–Schwarz and (H4) then yield

$$d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\rho))^2 \leq |\rho - \rho_0| \int_{\rho_0}^{\rho} |\dot{\gamma}|^2 ds \leq L^* \cdot C'\delta.$$

Step 3: triangle inequality. Squaring $d_{\mathcal{X}}(\gamma(\rho), B_1) \leq d_{\mathcal{X}}(\gamma(\rho), \gamma(\rho_0)) + d_{\mathcal{X}}(\gamma(\rho_0), B_1)$ with $(a + b)^2 \leq 2a^2 + 2b^2$ gives, uniformly for $\rho \in J^*$,

$$d_{\mathcal{X}}(\gamma(\rho), B_1)^2 \leq 2L^*C'\delta + 2K_1\delta = C_*\delta.$$

The endpoint case $\rho = \rho_\delta$ is included since $\rho_\delta \in J^*$. $\square$

Remark 5.2 (Why two inputs are needed: the D_H/D_I pairing). The proof is the 1D Sobolev embedding $H^1(J^*) \hookrightarrow L^\infty(J^*)$ applied to $\rho \mapsto d_{\mathcal{X}}(\gamma(\rho), B_1)$, decomposed into its two ingredients:

(i) the *distance term* (H1, $L^2(d\mu)$) encodes the integrated isoperimetric defect D_I via the Fusco–Julin estimate (8): many levels are close to balls;

(ii) the *kinetic term* (H4, $L^2(d\rho)$) encodes the integrated Cauchy–Schwarz defect D_H via the metric first variation of [2]: the family $\rho \mapsto F_\rho$ does not “curve away” from its trajectory faster than $\sqrt{\delta}$ in L^2 .

Step 1 (Markov) uses only (i) and extracts a single ρ_0 where $d^2 \leq K_1\delta$; it cannot place ρ_0 at ρ_δ . Step 2 (kinetic Cauchy–Schwarz) uses only (ii) and propagates the bound from ρ_0 to any $\rho \in J^*$ at cost $|\rho - \rho_0| \cdot C'\delta \leq L^*C'\delta$.

Either input alone fails. If one tried to extract the endpoint bound by Markov on a window of size ϵ near ρ_δ (to keep the boundary-layer transfer cost in [4] of order ϵ), the Markov constant becomes $4C/(c\epsilon)$, forcing the bound $d^2(\gamma(\rho_\delta), B_1) \lesssim \delta/\epsilon + \epsilon^2$. Optimising $\epsilon \sim \delta^{1/3}$ yields the lossy exponent $\delta^{2/3}$ instead of δ . Conversely, kinetic Cauchy–Schwarz alone gives only $\sqrt{\delta}$ -Hölder regularity of $\rho \mapsto d^2$, which without the integrated $L^2(d\mu)$ control of the distance term is vacuous as a pointwise bound.

The combination on the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$ yields the sharp δ exponent at the endpoint ρ_δ , which is exactly the upper boundary of the deficit interval and the location where the boundary-layer transfer of [4] has thickness $1 - \rho_\delta \asymp \sqrt{\delta}$.

Verification of hypotheses for the torsion application:

- (H1) is Theorem 1.1.
- (H2) follows from Hypothesis ??: on $G_{\tau_0} := \{\rho : w(\rho) \geq \tau_0\}$, with $\tau_0 = \rho_*/(2n)$, we have $\mu(d\rho) \geq \tau_0 d\rho$, while $\mathcal{L}^1(B_{\tau_0}) \leq C\delta$. Hence

$$\mu([a, b]) \geq \tau_0(b - a) - \tau_0 \mathcal{L}^1(B_{\tau_0}) \geq \tau_0(b - a) - C_{\text{low}}\delta.$$

- (H3) is the Talenti upper-density part of Hypothesis ??, with $M = 1/n$.
- (H4) is Theorem 1.2.

Proof of Theorem 1.3. After fixing the constants in (H2), shrink δ_0 if necessary so that the smallness condition in Theorem 5.1 holds on the order-one window guaranteed by (?). Apply Theorem 5.1 to $\gamma = [F_\rho]$. $\square$

Corollary 5.3 (Pointwise asymmetry at ρ_δ). *Setting $\hat{\rho} := \rho_\delta$,*

$$|E_{\rho_\delta} \Delta B_{\rho_\delta}|^2 \leq C(n, \rho_*, R) \delta_T(\Omega).$$

Proof. By the rescaling $F_\rho = \rho^{-1}E_\rho$ recorded at (13), the endpoint bound (4) yields $|E_{\rho_\delta} \Delta B_{\rho_\delta}|^2 \leq C(n, \rho_*, R) \rho_\delta^{2n} d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C(n, \rho_*, R) \delta_T(\Omega)$. The translation realising the infimum in $d_{\mathcal{X}}$ provides the centring required by the Fraenkel asymmetry on the right-hand side. $\square$

6 Conditionality

- Theorem 1.1: consumes the exact level-set deficit budget (7) from [1], the same-centre FJ trace package Hypothesis ??, the good-level metric upper gradient Hypothesis ??, the self-truncated velocity estimate of [5] on G_I only, and the bad-isoperimetric-level kinetic input (?).

- Theorem 1.2: consumes Theorem 1.1 on G_{τ_0} and the bad-density kinetic input (??). It does not use the old A0 assertion.
- Theorem 1.3: consumes the cleaned hypotheses (H1)–(H4). In the torsion application, (H2) and (H3) are supplied by Hypothesis ??.

Thus this block no longer depends on the centroid integrated Π -control input isolated in [3], and it no longer depends on the pointwise lower-gradient hypothesis Hyp-G. The low-gradient part of an isoperimetric-good level surface is handled by truncation and paid for by D_H . Pointwise (R)/(1.2) is never invoked as a proved input; the unresolved same-centre FJ/homothetic trace bridge is isolated in Hypothesis ??.

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